

$$\begin{array}{ll} m \text{ cte} & K/m = \frac{N-s^2}{m} \\ C \text{ cte} & N-s/m \\ K \text{ cte} & N/m \end{array}$$

$$m\ddot{u} + c\dot{u} + ku = 0$$

$\div m$

$$\ddot{u} = cm/s^2 = m/s^2$$

$$\dot{u} = cm/s = m/s$$

$$u = cm = m$$

$$\omega_n = s^{-1}, \text{ Hz}$$

$$\xi = \text{adimensional}$$

$$\ddot{u} + \frac{C}{m} \dot{u} + \frac{K}{m} u = 0$$

$$\ddot{u} + \frac{C}{m} \dot{u} + \omega_n^2 u = 0$$

$$\ddot{u} + \frac{C_c}{m} \dot{u} + \omega_n^2 u = 0$$

$$\ddot{u} + \xi \frac{(2m\omega_n)}{m} \dot{u} + \omega_n^2 u = 0$$

$$\ddot{u} + 2\xi\omega_n \dot{u} + \omega_n^2 u = 0$$

$$M^2 + 2\xi\omega_n M + \omega_n^2 M^0 = 0$$

- m_1 y m_2 reales y distintas ($b^2 - 4ac > 0$),
- m_1 y m_2 reales e iguales ($b^2 - 4ac = 0$), y
- m_1 y m_2 números conjugados complejos ($b^2 - 4ac < 0$).

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$M = \frac{-2\xi\omega_n \pm \sqrt{(2\xi\omega_n)^2 - 4\omega_n^2}}{2 \pm 1}$$

$$M = \frac{-2\xi\omega_n \pm \sqrt{4\xi^2\omega_n^2 - 4\omega_n^2}}{2}$$

$$M = \frac{-2\xi\omega_n \pm \sqrt{4\omega_n^2(\xi^2 - 1)}}{2}$$

$$M = \frac{-2\xi\omega_n \pm \sqrt{4\omega_n^2} \sqrt{\xi^2 - 1}}{2}$$

$$M = \frac{-\cancel{2}\xi\omega_n \pm \cancel{2}\omega_n \sqrt{\xi^2 - 1}}{2}$$

$$M = -\xi\omega_n \pm \omega_n \sqrt{\xi^2 - 1}$$

$$\sqrt{\xi^2 - 1} \quad \xi = 1$$

$$\sqrt{0} \quad \xi = \frac{C}{C_c} = 1$$

Amortiguamiento Crítico

1) Amortiguamiento Crítico $\xi = 1$

$$M = -\omega_n \pm \omega_n \sqrt{1-1}$$

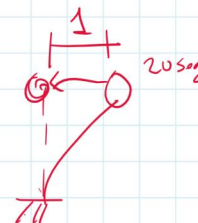
$$M = -\omega_n \quad M_1 = M_2 = -\omega_n$$

$$y = c_1 e^{m_1 x} + c_2 x e^{m_1 x}$$

(6)

$$u(t) = A e^{-\omega_n t} + B t e^{-\omega_n t}$$

$$u(t) = (A + Bt) e^{-\omega_n t}$$



$$\sqrt{\xi^2 - 1}$$

$$\xi > 1$$

Sobre amortiguado

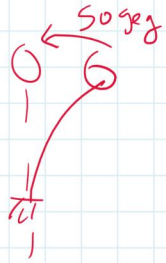
2) Sobre amortiguado $\xi > 1$ $M = -\xi \omega_n \pm \omega_n \sqrt{\xi^2 - 1}$

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

$$M_1 = -\xi \omega_n + \omega_n \sqrt{\xi^2 - 1}$$

$$M_2 = -\xi \omega_n - \omega_n \sqrt{\xi^2 - 1}$$

$$u(t) = A e^{(-\xi \omega_n + \omega_n \sqrt{\xi^2 - 1})t} + B e^{(-\xi \omega_n - \omega_n \sqrt{\xi^2 - 1})t}$$



3) Sub amortiguado

$$\sqrt{\xi^2 - 1}$$

$$0 < \xi < 1$$

$$M = -\xi \omega_n \pm \omega_n \sqrt{\xi^2 - 1}$$

$$M = -\xi \omega_n \pm \omega_n \sqrt{(-1)(-1)(\xi^2 - 1)}$$

$$M = -\xi \omega_n \pm \omega_n \sqrt{(-1)(1 - \xi^2)}$$

$$M = -\xi \omega_n \pm \omega_n \sqrt{-1} \sqrt{1 - \xi^2}$$

$$M = -\xi \omega_n \pm \omega_n \sqrt{1 - \xi^2} i$$

$$M = -\xi \omega_n \pm \omega_d i$$



$$\omega_d = \omega_n \sqrt{1 - \xi^2}$$

Frecuencia Natural amortiguada

$$y = c_1 e^{\alpha x} \cos \beta x + c_2 e^{\alpha x} \sin \beta x = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

$$m_1 = \alpha + i\beta \text{ y } m_2 = \alpha - i\beta$$

$$u(t) = e^{-\xi \omega_n t} (A \cos \omega_d t + B \sin \omega_d t)$$

Tarea:

$$u(0) = u_0$$

$$\dot{u}(0) = v_0$$

$$A = ? \quad B = ?$$